

Intel Corp.

Dividend Cut Expected - Our Thoughts On Other Items

What happened?

As we previewed over the weekend (see [here](#)) and widely expected by most investors, INTC finally announced that it has reduced its quarterly dividend by ~66% to \$0.125/share (\$0.50 annualized) from \$0.365/share (\$1.46 annualized) taking effect starting in C2Q:23. In addition to announcing the cut, the company reaffirmed FQ1:23 (Mar) guidance, generally reaffirmed FCF commentary for this year ("toward" the model of FCF Neutral in 2H:23), refuted [recent speculation](#) that its PC product roadmap has been negatively impacted by delays in 3nm orders with foundry partners, and fine tuned net capex target this year to "low 30%" of revenue. Lastly, it reiterated that it remains on track to deliver five nodes in four years, with Meteor Lake and Emerald Rapids still scheduled for launch in 2023 and Granite Rapids and Sierra Forest scheduled for 2024.

Our thoughts

We would highlight: 1) INTC's new dividend yield of ~1.9% is largely in-line with the ~1.7% average for the S&P500 - while potentially making the stock less attractive to dividend/income investors, this magnitude of cut was generally expected and consistent w/INTC's recent messaging; 2) although clearly revenue dependent, net capex commentary combined with the reiteration that 3rd party offsets will be 20-30% of gross capex implies the net capex of ~\$16B and gross capex of ~\$22B in C2023 - or largely in-line w/our model; 3) a return to mid 30%'s net capex to sales in 2024 would imply net capex in the ~\$20-21B range and, assuming 3rd party offsets step up to ~\$8-9B, implies gross capex could grow by as much as 30% next year; 4) we would caution reading too much into any delays in 3nm orders at TSMC as this would not necessarily drive any product delays given chiplet design and Intel already having 5nm graphic tile; 5) the company notably made no mention of the TSEM deal which it has previously indicated it is "working to get closed" in CQ1 (small as this is, we consider it key to the foundry strategy); and 6) with the stock having drifted another ~15% lower over the past 3mos (SOX +7%), it now trades at ~1.26x TBV if we adjust entirely for its MBLY stake, MBLY's book value included in INTC's book value, and INTC's intangibles (see Figure 5) - while downside to TBV is possible as INTC continues to burn cash net of 3rd party contributions, this seems unsustainably low given foundry comps alone (see Figure 6) as capacity utilization comes back in 2H and the company moves more clearly toward a separation of a fabless product group and a foundry manufacturing asset.

Valuation: PT \$29 (unchanged)

Highlights (US\$m)	12/20	12/21	12/22	12/23E	12/24E	12/25E	12/26E	12/27E
Revenues	77,867	74,718	63,054	50,871	61,292	70,472	82,832	99,933
EBIT (UBS)	25,292	22,205	7,915	3,083	12,013	17,345	24,162	34,285
Net earnings (UBS)	22,430	22,357	7,593	2,749	10,263	14,736	20,584	29,372
EPS (UBS, diluted) (US\$)	5.30	5.47	1.84	0.66	2.49	3.62	5.10	7.39
DPS (net) (US\$)	1.32	1.38	1.46	0.78	0.75	1.20	2.00	2.40
Net (debt) / cash	(12,506)	(9,688)	(13,713)	(20,189)	(23,828)	(20,654)	(14,145)	(1,352)
Profitability/valuation	12/20	12/21	12/22	12/23E	12/24E	12/25E	12/26E	12/27E
EBIT (UBS) margin %	32.5	29.7	12.6	6.1	19.6	24.6	29.2	34.3
ROIC (EBIT) %	29.3	24.1	7.6	2.7	9.5	12.5	16.2	21.5
EV/EBITDA (UBS core) x	6.5	6.8	7.9	7.3	4.6	3.7	3.0	2.3
P/E (UBS, diluted) x	10.4	10.2	20.9	39.3	10.5	7.2	5.1	3.5
Equity FCF (UBS) yield %	9.0	4.9	(6.0)	(10.8)	(7.8)	(0.9)	4.7	13.3
Dividend yield (net) %	2.4	2.5	3.8	3.0	2.9	4.6	7.7	9.2

Source: Company accounts, Thomson Reuters, UBS estimates. Metrics marked as (UBS) have had analyst adjustments applied. Valuations: based on an average share price that year, (E): based on a share price of US\$ 26.06 on 21-Feb-2023

Equities

Americas
Semiconductors

12-month rating **Neutral**

12m price target **US\$29.00**

Price (21 Feb 2023) **US\$26.06**

RIC: INTC.O BBG: INTC US

Trading data and key metrics

52-wk range	US\$52.25-25.04
Market cap.	US\$108b
Shares o/s	4,127m (COM)
Free float	100%
Avg. daily volume ('000)	38,647
Avg. daily value (m)	US\$1,101
Common s/h equity (12/23E)	US\$104b
P/BV (12/23E)	1.0x
Net debt to EBITDA (12/23E)	1.2x

EPS (UBS, diluted) (US\$)

	12/23E		% ch	Cons.
	From	To		
Q1E	(0.16)	(0.16)	NM	(0.15)
Q2E	(0.09)	(0.09)	NM	0.02
Q3E	0.21	0.21	1	0.24
Q4E	0.70	0.70	0	0.41
12/23E	0.66	0.66	0	0.55
12/24E	2.46	2.49	1	1.88
12/25E	3.56	3.62	2	2.58

Timothy Arcuri

Analyst

timothy.arcuri@ubs.com

+1-415-352 5676

Jason Park

Associate Analyst

jason.s.park@ubs.com

+1-212-713 8728

Aman Gulani

Associate Analyst

aman.gulani@ubs.com

+1-212-713-2000

This report has been prepared by UBS Securities LLC. **ANALYST CERTIFICATION AND REQUIRED DISCLOSURES, INCLUDING INFORMATION ON THE QUANTITATIVE RESEARCH REVIEW PUBLISHED BY UBS, BEGIN ON PAGE 11.** UBS does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.

Our price target of \$29 (unchanged) is based on the same valuation method where the EV/FCF NTM multiple of ~17x is applied to the largely unchanged normalized FCF base of ~\$8.2B (from ~\$8.1B), which is calculated by allocating a ~20% weighting to the average of CY23/CY24E FCF of -\$2.7B (from -\$2.4B) and ~80% weight to the average of CY25/CY26E FCF of ~\$11.0B (from ~\$10.7B). Ultimately we think a separation of the company will drive value and we are reasonably optimistic in the manufacturing roadmap but the payoff remains too far out to get excited at this point.

Estimate Changes

Our FQ1:23E Revs/EPS remain unchanged at \$11.0B/(\$0.16) and we model CY23/24E Revs/EPS of \$50.9B/\$0.66 (unchanged) and \$61.3B/\$2.49 (from \$61.3B/\$2.46).

Figure 1: New vs Old Estimates

	C2023E			C2024E		
	New	Old	Change	New	Old	Change
Revenue (\$MM)	\$50,871	\$50,871	0.0%	\$61,292	\$61,292	0.0%
Y/Y Growth	-19.3%	-19.3%		20.5%	20.5%	
Gross Margin	42.2%	42.2%	0.0%	47.1%	47.1%	0.0%
OpEx % of Revs	36.2%	36.2%	0.0%	27.5%	27.5%	0.0%
Operating Margin	6.1%	6.1%	0.0%	19.6%	19.6%	0.0%
EPS	\$0.66	\$0.66	0.0%	\$2.49	\$2.46	1.1%

Source: Company reports, UBS estimates

Figure 2: UBSe vs. Street

	Actual 2019	Actual 2020	Actual 2021	Actual 2022	Estimate 2023/1F	Estimate 2023/2F	Estimate 2023/3F	Estimate 2023/4F	Estimate 2023	Estimate 2024/1F	Estimate 2024/2F	Estimate 2024/3F	Estimate 2024/4F	Estimate 2024	CY2019	CY2020	CY2021	CY2022	CY2023	CY2024
Revenues (\$MM)																				
UBS	\$71,965	\$77,867	\$74,718	\$63,054	\$10,976	\$11,222	\$12,926	\$15,746	\$50,871	\$15,037	\$14,798	\$15,422	\$16,035	\$61,292	\$71,965	\$77,867	\$74,718	\$63,054	\$50,871	\$61,292
UBS Old					\$10,976	\$11,222	\$12,926	\$15,746	\$50,871	\$15,037	\$14,798	\$15,422	\$16,035	\$61,292					\$50,871	\$61,292
Consensus					\$11,118	\$11,761	\$13,256	\$14,575	\$50,711	\$13,818	\$14,200	\$14,976	\$15,560	\$58,553					\$50,711	\$58,553
Delta					(1.3%)	(4.6%)	(2.5%)	8.0%	0.3%	8.8%	4.2%	3.0%	3.1%	4.7%					0.3%	4.7%
EPS																				
UBS (Non-GAAP)	\$4.87	\$5.30	\$5.47	\$1.84	-\$0.16	-\$0.09	\$0.21	\$0.70	\$0.66	\$0.62	\$0.57	\$0.61	\$0.68	\$2.49	\$4.87	\$5.30	\$5.47	\$1.84	\$0.66	\$2.49
UBS (Non-GAAP) Old					-\$0.16	-\$0.09	\$0.21	\$0.70	\$0.66	\$0.62	\$0.57	\$0.60	\$0.67	\$2.46					\$0.66	\$2.46
Consensus (Non-GAAP)					-\$0.16	\$0.00	\$0.22	\$0.38	\$0.45	\$0.33	\$0.39	\$0.49	\$0.56	\$1.78					\$0.45	\$1.78
Delta					2.9%	(2838%)	(3.9%)	83.1%	47.6%	87.1%	46.1%	25.3%	21.4%	40.2%					47.6%	40.2%

Source: Factset, Company reports, UBS estimates

Figure 3: Summary model

Intel (INTC) - Key Metrics (in \$MM except EPS and units)	F2018A C2018A	F2019A C2019A	F2020A C2020A	F2021A C2021A	F2022A C2022A	Estimates FQ1E CQ1E	FQ2E CQ2E	FQ3E CQ3E	FQ4E CQ4E	F2023E C2023E	FQ1E CQ1E	FQ2E CQ2E	FQ3E CQ3E	FQ4E CQ4E	F2024E C2024E
Revenue (GAAP)	70,848	71,965	77,867	79,024	83,054	10,976	11,222	12,926	15,746	50,871	15,037	14,798	15,422	16,035	61,292
(-) NSG	--	--	4,960	3,897	--	--	--	--	--	--	--	--	--	--	--
Revenue (Non GAAP)	\$70,848	\$71,965	\$77,867	\$74,718	\$83,054	\$10,976	\$11,222	\$12,926	\$15,746	\$50,871	\$15,037	\$14,798	\$15,422	\$16,035	\$61,292
Y/Y Growth	13%	2%	8%	(4%)	(16%)	(40%)	(27%)	(16%)	12%	(19%)	37%	32%	19%	2%	20%
Gross Profit (Non-GAAP)	44,842	43,264	44,823	43,103	29,794	4,281	4,377	5,429	7,401	21,487	7,067	6,955	7,248	7,616	28,887
Gross Margin	63.3%	60.1%	57.6%	57.7%	47.3%	39.0%	39.0%	42.0%	47.0%	42.2%	47.0%	47.0%	47.0%	47.5%	47.1%
R&D (Non-GAAP)	13,543	13,362	13,556	15,190	17,528	3,900	3,650	3,400	3,100	14,050	3,100	3,300	3,400	3,500	13,300
M&A (Non-GAAP)	6,150	6,150	8,180	6,543	7,002	1,600	1,500	1,400	1,400	5,900	1,400	1,300	1,300	1,250	5,250
Other	259	--	(155)	(835)	--	--	--	--	--	--	--	--	--	--	--
Total Opex (Non-GAAP)	\$20,552	\$19,512	\$19,581	\$20,898	\$21,879	\$5,130	\$4,813	\$4,422	\$4,039	\$18,404	\$4,060	\$4,189	\$4,285	\$4,329	\$16,874
Y/Y Growth	0%	(5%)	0%	7%	5%	(7%)	(12%)	(18%)	(27%)	(16%)	(21%)	(13%)	(3%)	7%	(8%)
Operating Income (Non-GAAP)	24,290	23,752	25,292	22,205	7,915	(849)	(437)	1,007	3,361	3,083	3,007	2,766	2,953	3,287	12,013
Operating Margin	34.3%	33.0%	32.5%	29.7%	12.6%	(7.7%)	(3.9%)	7.8%	21.3%	6.1%	20.0%	18.7%	19.1%	20.5%	19.6%
Non-GAAP EPS	\$4.58	\$4.87	\$5.30	\$5.47	\$1.84	(\$0.16)	(\$0.09)	\$0.21	\$0.70	\$0.66	\$0.62	\$0.57	\$0.61	\$0.68	\$2.49
Free Cash Flow															
CFFO	29,432	33,145	35,384	29,991	15,433	(4,080)	1,907	4,521	8,056	10,404	2,837	4,643	5,327	5,702	18,609
Less: Gross Capex	15,181	16,213	14,259	18,733	24,844	5,500	5,500	5,500	5,500	22,000	5,500	6,500	7,500	7,500	27,000
Add: Contribution from Partners	--	--	--	--	874	1,498	1,498	1,503	1,503	6,000	1,988	2,113	2,238	2,238	8,575
Net Capex	15,181	16,213	14,259	18,733	23,970	4,003	4,003	3,998	3,998	16,000	3,513	4,388	5,263	5,263	18,425
FCF	\$14,251	\$16,932	\$21,125	\$11,258	(\$8,537)	(\$8,083)	(\$2,096)	\$524	\$4,059	(\$5,596)	(\$575)	\$255	\$64	\$440	\$184
% of revenue	20%	24%	27%	15%	(14%)	(74%)	(19%)	4%	26%	(11%)	(4%)	2%	0%	3%	0%
Y/Y Growth	38%	19%	22%	(47%)	(17%)	(63%)	(67%)	(109%)	41%	(34%)	(33%)	(112%)	(88%)	(89%)	--
Product Revenue															
Client Computing (CCG)	--	\$37,938	\$40,535	\$41,067	\$31,708	\$4,527	\$5,021	\$6,394	\$8,769	\$24,711	\$7,973	\$7,404	\$7,644	\$7,892	\$30,913
Datacenter and AI (DCAI)	--	\$21,696	\$23,413	\$22,691	\$19,196	\$3,080	\$3,496	\$3,741	\$4,062	\$14,919	\$3,862	\$4,042	\$4,123	\$4,285	\$16,415
Network and Edge (NEX)	--	\$6,829	\$7,132	\$7,976	\$8,873	\$1,855	\$1,762	\$1,762	\$1,850	\$7,229	\$1,887	\$2,038	\$2,242	\$2,332	\$8,499
Accelerated Computing Systems and Graphics (AXG)	--	\$606	\$651	\$774	\$837	\$247	\$264	\$283	\$303	\$1,097	\$339	\$380	\$425	\$476	\$1,620
Mobileye	--	\$879	\$967	\$1,386	\$1,869	\$565	\$576	\$622	\$660	\$2,423	\$706	\$755	\$801	\$849	\$3,111
Intel Foundry Services (IFS)	--	\$461	\$715	\$786	\$895	\$223	\$223	\$246	\$282	\$975	\$291	\$300	\$309	\$318	\$1,217
All Other	--	\$4,150	\$5,091	\$5,019	\$196	\$30	\$30	\$30	\$30	\$120	\$30	\$30	\$30	\$30	\$120
Operating Margins															
Client Computing (CCG)	--	43%	39%	38%	20%	--	--	--	--	--	--	--	--	--	--
Datacenter and AI (DCAI)	--	46%	47%	37%	12%	--	--	--	--	--	--	--	--	--	--
Network and Edge (NEX)	--	25%	12%	21%	8%	--	--	--	--	--	--	--	--	--	--
Accelerated Computing Systems and Graphics (AXG)	--	(58%)	(62%)	(156%)	(205%)	--	--	--	--	--	--	--	--	--	--
Mobileye	--	36%	33%	40%	37%	--	--	--	--	--	--	--	--	--	--
Intel Foundry Services (IFS)	--	(46%)	6%	(3%)	(36%)	--	--	--	--	--	--	--	--	--	--
All Other	--	(134%)	(79%)	(114%)	(2,864%)	--	--	--	--	--	--	--	--	--	--
Q/Q Growth (Top Segment Level)															
Client Computing (CCG)	--	--	--	--	--	(32%)	11%	27%	37%	--	(9%)	(7%)	3%	3%	--
Datacenter and AI (DCAI)	--	--	--	--	--	(15%)	(5%)	7%	7%	--	(1%)	2%	2%	4%	--
Network and Edge (NEX)	--	--	--	--	--	(10%)	(5%)	--	5%	--	2%	6%	10%	4%	--
Accelerated Computing Systems and Graphics (AXG)	--	--	--	--	--	--	7%	7%	7%	--	12%	12%	12%	12%	--
Mobileye	--	--	--	--	--	--	2%	8%	6%	--	7%	7%	6%	6%	--
Intel Foundry Services (IFS)	--	--	--	--	--	(30%)	--	10%	15%	--	3%	3%	3%	3%	--
All Other	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Y/Y Growth (Top Segment Level)															
Client Computing (CCG)	--	--	7%	1%	(23%)	(51%)	(34%)	(21%)	32%	(22%)	76%	47%	20%	(10%)	25%
Datacenter and AI (DCAI)	--	--	8%	(3%)	(15%)	(39%)	(25%)	(11%)	(7%)	(22%)	8%	16%	10%	7%	10%
Network and Edge (NEX)	--	--	4%	12%	11%	(16%)	(24%)	(22%)	(10%)	(19%)	2%	16%	27%	26%	18%
Accelerated Computing Systems and Graphics (AXG)	--	--	7%	15%	8%	13%	42%	53%	23%	31%	37%	44%	50%	57%	48%
Mobileye	--	--	10%	43%	35%	43%	25%	38%	17%	30%	25%	31%	29%	29%	28%
Intel Foundry Services (IFS)	--	--	55%	10%	14%	(21%)	83%	44%	(11%)	9%	30%	34%	26%	13%	25%
All Other	--	--	23%	(1%)	(96%)	(55%)	(6%)	(55%)	--	(39%)	--	--	--	--	--

Source: Company reports, UBS estimates

NOTE: FCF embeds the following assumed offsets to gross capex over the longer-term: 1) ITC - we assume half of INTC capex is domestic and take a 25% credit against that, 2) US CHIPS Act subsidy, 3) European equivalent subsidy, 4) Brookfield AZ fab funding

Valuation

Our price target of \$29 (unchanged) is based on the same valuation method where the EV/FCF NTM multiple of ~17x is applied to the largely unchanged normalized FCF base of ~\$8.2B (from ~\$8.1B), which is calculated by allocating a ~20% weighting to the average of CY23/CY24E FCF of -\$2.7B (from -\$2.4B) and ~80% weight to the average of CY25/CY26E FCF of ~\$11.0B (from ~\$10.7B). Ultimately we think a separation of the company will drive value and we are reasonably optimistic in the manufacturing roadmap but the payoff remains too far out to get excited at this point.

Figure 4: Valuation Methodology

Valuation	New	Old
FCF		
C2023E	\$ (5,596)	\$ (4,684)
C2024E	\$ 184	\$ (40)
Avg of CY23/24E	\$ (2,706)	\$ (2,362)
Weight	20%	20%
C2025E	\$ 7,932	\$ 7,608
C2026E	\$ 14,130	\$ 13,762
Avg of CY25/26E	\$ 11,031	\$ 10,685
Weight	80%	80%
Normalized FCF Base	\$ 8,216	\$ 8,075
EV/FCF Multiple	17x	17x
Enterprise Value	\$ 135,569	\$ 137,280
Less: Debt	\$ 41,651	\$ 41,651
Add: Cash	\$ 27,374	\$ 24,866
Equity Value	121,292	120,496
Shares Outstanding	4,143	4,143
Equity Value/share	\$29	\$29

Source: UBS estimates, Company reports

Figure 5: INTC Adjusted P/TBV for MBLY

MBLY	\$41.07
Shares Outstanding	801.91
MBLY Equity Value	\$32,934.44
INTC Ownership (94%)	94%
INTC Equity Stake	\$30,958.38
INTC	\$26.06
Shares Outstanding	4,133
INTC Equity Value	\$107,705.98
INTC Equity Value Adj for MBLY	\$76,747.60
INTC Share Price Adj. for MBLY	\$18.57
INTC TBV/Sh	\$16.39
MBLY TBV/Sh	\$1.72
INTC TBV/Sh Adj for MBLY at 94%	\$14.77
INTC P/TBV	1.59x
INTC P/TBV Adj. for MBLY	1.26x

Source: Factset, UBSe

Figure 6: Foundry Comps P/TBV

Ticker	Company Name	P/TBV
2330-TW	TAIWAN SEMICON MAN	4.58x
UMC-US	UNITED MICROELECTRONICS CORP	2.05x
5347-TW	VANGUARD INTL SEMI	3.45x
GFS-US	GLOBALFOUNDRIES INC	3.56x
TSEM-US	TOWER SEMICONDUCTOR LTD	2.39x
MBLY	MOBILEYE GLOBAL INC	23.84x
INTC	INTEL CORP	1.59x
INTC Adj. for MBLY	Intel P/TBV Adj. for MBLY	1.26x

Source: Factset, UBSe

Intel Corp. (INTC.O)

	12/20	12/21	12/22	12/23E	%ch	12/24E	%ch	12/25E	12/26E	12/27E
Income Statement (US\$m)										
Revenues	77,867	74,718	63,054	50,871	-19.3	61,292	20.5	70,472	82,832	99,933
Gross profit	44,823	43,103	29,794	21,487	-27.9	28,887	34.4	35,260	43,298	54,237
EBITDA (UBS)	37,531	33,997	20,950	16,608	-20.7	26,150	57.5	32,507	40,291	51,331
Depreciation & amortisation	(12,239)	(11,792)	(13,035)	(13,525)	-3.8	(14,137)	-4.5	(15,161)	(16,129)	(17,046)
EBIT (UBS)	25,292	22,205	7,915	3,083	-61.1	12,013	289.7	17,345	24,162	34,285
Associates & investment income	0	0	0	0	-	0	-	0	0	0
Other non-operating income	0	0	0	0	-	0	-	0	0	0
Net interest	1,423	2,247	(1)	77	-	61	-20.1	(9)	55	271
Exceptionals (incl goodwill)	0	0	5,435	0	-	0	-	0	0	0
Pre-tax profit	26,715	24,452	13,349	3,159	-76.3	12,075	282.2	17,337	24,217	34,556
Tax	(4,285)	(2,095)	(321)	(411)	-27.9	(1,811)	-341.0	(2,600)	(3,632)	(5,183)
Profit after tax	22,430	22,357	13,028	2,749	-78.9	10,263	273.4	14,736	20,584	29,372
Preference dividends	0	0	0	0	-	0	-	0	0	0
Minorities	0	0	0	0	-	0	-	0	0	0
Extraordinary items	(1,531)	(2,489)	(5,011)	(2,241)	55.3	(2,375)	-5.9	(2,246)	(2,640)	(3,185)
Net earnings (local GAAP)	20,899	19,868	8,017	507	-93.7	7,889	NM	12,490	17,944	26,187
Net earnings (UBS)	22,430	22,357	7,593	2,749	-63.8	10,263	273.4	14,736	20,584	29,372
Tax rate (%)	16.0	8.6	2.4	13.0	NM	15.0	15.4	15.0	15.0	15.0
Per Share (US\$)										
EPS (UBS, diluted)	5.30	5.47	1.84	0.66	-64.0	2.49	275.7	3.62	5.10	7.39
EPS (local GAAP, diluted)	4.94	4.86	1.95	0.12	-93.7	1.92	NM	3.07	4.44	6.59
EPS (UBS, basic)	5.34	5.51	1.85	0.66	-64.1	2.50	276.1	3.63	5.12	7.43
DPS (net) (US\$)	1.32	1.38	1.46	0.78	-46.8	0.75	-3.2	1.20	2.00	2.40
Cash EPS (UBS, diluted) ¹	8.19	8.35	5.01	3.93	-21.6	5.92	50.9	7.34	9.09	11.69
Book value per share	19.79	23.49	24.58	25.00	1.7	27.91	11.6	31.61	35.72	41.17
Average shares (diluted)	4,232	4,090	4,116	4,143	0.7	4,118	-0.6	4,074	4,037	3,972
Balance Sheet (US\$m)										
Cash and equivalents	23,895	28,413	28,338	21,462	-24.3	15,973	-25.6	15,397	20,906	30,449
Other current assets	23,354	29,305	22,069	21,298	-3.5	22,114	3.8	22,644	24,054	27,422
Total current assets	47,249	57,718	50,407	42,761	-15.2	38,087	-10.9	38,041	44,960	57,871
Net tangible fixed assets	56,584	63,245	80,860	89,335	10.5	102,198	14.4	117,037	131,907	145,862
Net intangible fixed assets	26,971	26,963	27,591	27,591	0.0	27,591	0.0	27,591	27,591	27,591
Investments / other assets	22,287	20,480	23,245	23,245	0.0	23,245	0.0	23,245	23,245	23,245
Total assets	153,091	168,406	182,103	182,932	0.5	191,121	4.5	205,914	227,704	254,569
Trade payables & other ST liabilities	22,250	22,871	27,788	25,567	-8.0	24,970	-2.3	28,956	34,102	40,813
Short term debt	2,504	4,591	4,367	1,850	-57.6	3,750	102.7	1,000	3,250	0
Total current liabilities	24,754	27,462	32,155	27,417	-14.7	28,720	4.8	29,956	37,352	40,813
Long term debt	33,897	33,510	37,684	39,801	5.6	36,051	-9.4	35,051	31,801	31,801
Other long term liabilities	13,402	12,043	8,978	10,233	14.0	10,315	0.8	11,145	12,169	13,512
Preferred shares	0	0	0	0	-	0	-	0	0	0
Total liabilities (incl pref shares)	72,053	73,015	78,817	77,451	-1.7	75,086	-3.1	76,152	81,322	86,126
Common s/h equity	81,038	95,391	101,423	103,618	2.2	114,172	10.2	127,899	144,519	166,580
Minority interests	0	0	1,863	1,863	0.0	1,863	0.0	1,863	1,863	1,863
Total liabilities & equity	153,091	168,406	182,103	182,932	0.5	191,121	4.5	205,914	227,704	254,569
Cash Flow (US\$m)										
Net income (before pref divs)	20,899	19,868	8,017	507	-93.7	7,889	NM	12,490	17,944	26,187
Depreciation & amortisation	12,239	11,792	13,035	13,525	3.8	14,137	4.5	15,161	16,129	17,046
Net change in working capital	2,179	(4,873)	(4,508)	(1,450)	67.8	(1,413)	2.6	3,456	3,736	3,343
Other operating	67	3,204	(1,111)	(2,178)	-96.0	(2,004)	8.0	(2,125)	(1,754)	(1,241)
Operating cash flow	35,384	29,991	15,433	10,404	-32.6	18,609	78.9	28,982	36,055	45,335
Tangible capital expenditure	(14,259)	(18,733)	(24,844)	(22,000)	11.4	(27,000)	-22.7	(30,000)	(31,000)	(31,000)
Intangible capital expenditure	0	0	0	0	-	0	-	0	0	0
Net (acquisitions) & disposals	(1,031)	(1,805)	0	0	-	0	-	0	0	0
Other investing	(5,506)	(4,629)	14,367	8,255	-42.5	6,082	-26.3	3,830	1,024	1,343
Investing cash flow	(20,796)	(25,167)	(10,477)	(13,745)	-31.2	(20,918)	-52.2	(26,170)	(29,976)	(29,657)
Equity dividends paid	(5,568)	(5,644)	(5,997)	(3,209)	46.5	(3,088)	3.8	(4,889)	(8,075)	(9,534)
Share issues / (buybacks)	(13,332)	(1,395)	977	1,075	10.0	(818)	-	(700)	(570)	(2,427)
Other financing	261	(1,297)	(2)	0	-	0	-	0	0	0
Change in debt & pref shares	5,560	2,474	5,509	(400)	-	(1,850)	-362.5	(3,750)	(1,000)	(3,250)
Financing cash flow	(13,079)	(5,862)	487	(2,535)	-	(5,756)	-127.1	(9,338)	(9,645)	(15,210)
Cash flow inc/(dec) in cash	1,509	(1,038)	5,443	(5,876)	-	(8,064)	-37.2	(6,526)	(3,565)	468
FX / non cash items	9,263	5,556	(5,518)	(1,000)	81.9	2,575	-	5,950	9,075	9,075
Balance sheet inc/(dec) in cash	10,772	4,518	(75)	(6,876)	NM	(5,489)	20.2	(576)	5,510	9,543

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts. ¹ Cash EPS (UBS, diluted) is calculated using UBS net income adding back depreciation and amortization.

Intel Corp. (INTC.O)

Valuation (x)	12/20	12/21	12/22	12/23E	12/24E	12/25E	12/26E	12/27E
P/E (local GAAP, diluted)	11.1	11.5	19.8	NM	13.6	8.5	5.9	4.0
P/E (UBS, diluted)	10.4	10.2	20.9	39.3	10.5	7.2	5.1	3.5
P/CEPS	6.6	6.6	7.7	6.6	4.4	3.5	2.9	2.2
Equity FCF (UBS) yield %	9.0	4.9	(6.0)	(10.8)	(7.8)	(0.9)	4.7	13.3
Dividend yield (net) %	2.4	2.5	3.8	3.0	2.9	4.6	7.7	9.2
P/BV	2.8	2.4	1.6	1.0	0.9	0.8	0.7	0.6
EV/revenues (core)	3.1	3.1	2.6	2.4	2.0	1.7	1.5	1.2
EV/EBITDA (UBS core)	6.5	6.8	7.9	7.3	4.6	3.7	3.0	2.3
EV/EBIT (core)	9.6	10.5	20.8	39.1	10.0	6.9	5.0	3.5
EV/OpFCF (core)	9.5	22.3	NM	NM	NM	20.2	9.2	5.1
EV/op. invested capital	2.8	2.5	1.6	1.0	0.9	0.9	0.8	0.8
Enterprise value (US\$m)	12/20	12/21	12/22	12/23E	12/24E	12/25E	12/26E	12/27E
Market cap.	235,781	228,132	157,547	107,550	107,550	107,550	107,550	107,550
Net debt (cash)	14,192	11,097	11,097	16,951	16,951	16,951	16,951	16,951
Buy out of minorities	0	0	1,863	1,863	1,863	1,863	1,863	1,863
Pension provisions/other	0	0	0	0	0	0	0	0
Total enterprise value	249,973	239,229	170,507	126,363	126,363	126,363	126,363	126,363
Non core assets	(7,344)	(7,138)	(5,912)	(5,912)	(5,912)	(5,912)	(5,912)	(5,912)
Core enterprise value	242,629	232,091	164,595	120,451	120,451	120,451	120,451	120,451
Growth (%)	12/20	12/21	12/22	12/23E	12/24E	12/25E	12/26E	12/27E
Revenue	8.2	(4.0)	(15.6)	(19.3)	20.5	15.0	17.5	20.6
EBITDA (UBS)	8.5	(9.4)	(38.4)	(20.7)	57.5	24.3	23.9	27.4
EBIT (UBS)	6.5	(12.2)	(64.4)	(61.1)	NM	44.4	39.3	41.9
EPS (UBS, diluted)	8.8	3.1	(66.3)	(64.0)	NM	45.1	40.9	45.0
Net DPS	4.8	4.8	5.4	(46.8)	(3.2)	60.0	66.7	20.0
Margins & Profitability (%)	12/20	12/21	12/22	12/23E	12/24E	12/25E	12/26E	12/27E
Gross profit margin	57.6	57.7	47.3	42.2	47.1	50.0	52.3	54.3
EBITDA margin	48.2	45.5	33.2	32.6	42.7	46.1	48.6	51.4
EBIT (UBS) margin	32.5	29.7	12.6	6.1	19.6	24.6	29.2	34.3
Net earnings (UBS) margin	28.8	29.9	12.0	5.4	16.7	20.9	24.9	29.4
ROIC (EBIT)	29.3	24.1	7.6	2.7	9.5	12.5	16.2	21.5
ROIC post tax	24.6	22.1	7.3	2.3	8.0	10.6	13.7	18.3
ROE (UBS)	28.3	25.3	7.7	2.7	9.4	12.2	15.1	18.9
Capital structure & Coverage (x)	12/20	12/21	12/22	12/23E	12/24E	12/25E	12/26E	12/27E
Net debt / EBITDA	0.3	0.3	0.7	1.2	0.9	0.6	0.4	0.0
Net debt / total equity %	15.4	10.2	13.3	19.1	20.5	15.9	9.7	0.8
Net debt / (net debt + total equity) %	13.4	9.2	11.7	16.1	17.0	13.7	8.8	0.8
Net debt/EV %	5.7	4.6	6.9	13.4	17.4	17.6	13.8	6.1
Capex / depreciation %	136.0	188.2	NM	189.1	NM	NM	NM	NM
Capex / revenue %	18.3	25.1	NM	NM	NM	NM	NM	NM
EBIT / net interest	-	-	NM	-	-	NM	-	-
Dividend cover (UBS)	4.0	4.0	1.3	0.9	3.3	3.0	2.6	3.1
Div. payout ratio (UBS) %	24.7	25.1	78.8	116.8	30.1	33.1	39.1	32.3
Revenues by division (US\$m)	12/20	12/21	12/22	12/23E	12/24E	12/25E	12/26E	12/27E
Others	77,867	74,718	63,054	50,871	61,292	70,472	82,832	99,933
Total	77,867	74,718	63,054	50,871	61,292	70,472	82,832	99,933
EBIT (UBS) by division (US\$m)	12/20	12/21	12/22	12/23E	12/24E	12/25E	12/26E	12/27E
Others	25,292	22,205	7,915	3,083	12,013	17,345	24,162	34,285
Total	25,292	22,205	7,915	3,083	12,013	17,345	24,162	34,285

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.

Forecast returns

Forecast price appreciation	+11.3%
Forecast dividend yield	2.0%
Forecast stock return	+13.3%
Market return assumption	9.7%
Forecast excess return	+3.6%

Company Description

Intel Corporation is the leading producer of x86-based microprocessor chips and platforms for client computing (~50% of sales), data center (30-35%), internet of things (~5%), memory solutions (~8%), security (~1%), and programmable solutions (~3%). Its products and solutions are widely used in notebook and desktop PCs, enterprise/cloud servers, communications infrastructure equipment, mobile phones and tablets, wireless and wired connectivity products, IoT, industrial, military, and automotive applications, among others.

Valuation Method and Risk Statement

INTC: Due to ramping depreciation, we focus on EV/FCF. NVDA has built a formidable moat for new compute-intensive workloads in the data center and could ultimately leverage its GPU architecture to more broadly displace INTC. New client and server CPUs from AMD also present a threat that we could be under-estimating. Lastly, its push into memory could continue to constrain FCF.

Quantitative Research Review

UBS publishes a quantitative assessment of its analysts' responses to certain questions about the likelihood of an occurrence of a number of short term factors in a product known as the 'Quantitative Research Review'. The views for this month can be found below. Views contained in this assessment on a particular stock reflect only the views on those short term factors which are a different timeframe to the 12-month timeframe reflected in any equity rating set out in this note. For previous responses please make reference to (i) previous UBS research reports; and (ii) where no applicable research report was published that month, the Quantitative Research Review which can be found at <https://neo.ubs.com/quantitative>, or contact your UBS sales representative for access to the report or the Quantitative Research Team on qa@ubs.com. A consolidated report which contains all responses is also available and again you should contact your UBS sales representative for details and pricing or the Quantitative Research Team on the email above.

Intel Corp.

Question	Response
1. Is the industry structure facing the firm likely to improve or deteriorate over the next year? Rate on a scale of 1-5 (1 = getting worse, 3 = no change, 5 = getting better, N/A = no view)	N/A
2. Is the regulatory/government environment facing the firm likely to improve or deteriorate over the next year? Rate on a scale of 1-5 (1 = getting tougher 3 = no change, 5 = getting better, N/A = no view)	N/A
3. Over the last 3-6 months in broad terms have things been improving/no change/getting worse for this stock? Rate on a scale of 1-5 (1 = getting a lot worse, 3 = not much change, 5 = getting a lot better, N/A = no view)	N/A
4. Relative to the current CONSENSUS EPS forecast, is the next company EPS update likely to lead to: (1 = negative surprise vs consensus, 3 = in-line with consensus, 5 = positive surprise vs consensus expectations, N/A = no view)	N/A
5. What's driving the difference?	
6. Relative to YOUR current earnings forecast, is there relatively greater risk at the next earnings result of: (1 = downside skew risk to earnings, 3 = equal upside or downside risk to earnings, 5 = upside skew risk to earnings, N/A = no view)	N/A
7. What's driving the difference?	
8. Is there an upcoming catalyst for the company over the next three months?	Negative Catalyst
9. Is there an actual or approximate date for the catalyst?	
10. Is the catalyst date an actual or approximate date?	N/A
11. What is the catalyst?	

Required Disclosures

This report has been prepared by UBS Securities LLC, an affiliate of UBS AG. UBS AG, its subsidiaries, branches and affiliates are referred to herein as UBS.

For information on the ways in which UBS manages conflicts and maintains independence of its research product; historical performance information; certain additional disclosures concerning UBS research recommendations; and terms and conditions for certain third party data used in research report, please visit <https://www.ubs.com/disclosures>. The figures contained in performance charts refer to the past; past performance is not a reliable indicator of future results. Additional information will be made available upon request. UBS Securities Co. Limited is licensed to conduct securities investment consultancy businesses by the China Securities Regulatory Commission. UBS acts or may act as principal in the debt securities (or in related derivatives) that may be the subject of this report. This recommendation was finalized on: 22 February 2023 07:56 PM GMT. UBS has designated certain Research department members as Derivatives Research Analysts where those department members publish research principally on the analysis of the price or market for a derivative, and provide information reasonably sufficient upon which to base a decision to enter into a derivatives transaction. Where Derivatives Research Analysts co-author research reports with Equity Research Analysts or Economists, the Derivatives Research Analyst is responsible for the derivatives investment views, forecasts, and/or recommendations. **Quantitative Research Review:** UBS publishes a quantitative assessment of its analysts' responses to certain questions about the likelihood of an occurrence of a number of short term factors in a product known as the 'Quantitative Research Review'. Views contained in this assessment on a particular stock reflect only the views on those short term factors which are a different timeframe to the 12-month timeframe reflected in any equity rating set out in this note. For the latest responses, please see the Quantitative Research Review Addendum at the back of this report, where applicable. For previous responses please make reference to (i) previous UBS research reports; and (ii) where no applicable research report was published that month, the Quantitative Research Review which can be found at <https://neo.ubs.com/quantitative>, or contact your UBS sales representative for access to the report or the Quantitative Research Team on qa@ubs.com. A consolidated report which contains all responses is also available and again you should contact your UBS sales representative for details and pricing or the Quantitative Research team on the email above.

Analyst Certification:

Each research analyst primarily responsible for the content of this research report, in whole or in part, certifies that with respect to each security or issuer that the analyst covered in this report: (1) all of the views expressed accurately reflect his or her personal views about those securities or issuers and were prepared in an independent manner, including with respect to UBS, and (2) no part of his or her compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed by that research analyst in the research report.

UBS Investment Research: Global Equity Rating Definitions

12-Month Rating	Definition	Coverage ¹	IB Services ²
Buy	FSR is > 6% above the MRA.	55%	24%
Neutral	FSR is between -6% and 6% of the MRA.	36%	21%
Sell	FSR is > 6% below the MRA.	9%	18%
Short-Term Rating	Definition	Coverage ³	IB Services ⁴
Buy	Stock price expected to rise within three months from the time the rating was assigned because of a specific catalyst or event.	<1%	<1%
Sell	Stock price expected to fall within three months from the time the rating was assigned because of a specific catalyst or event.	<1%	<1%

Source: UBS. Rating allocations are as of 31 December 2022.

1:Percentage of companies under coverage globally within the 12-month rating category.

2:Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

3:Percentage of companies under coverage globally within the Short-Term rating category.

4:Percentage of companies within the Short-Term rating category for which investment banking (IB) services were provided within the past 12 months.

KEY DEFINITIONS: Forecast Stock Return (FSR) is defined as expected percentage price appreciation plus gross dividend yield over the next 12 months. In some cases, this yield may be based on accrued dividends. **Market Return Assumption (MRA)** is defined as the one-year local market interest rate plus 5% (a proxy for, and not a forecast of, the equity risk premium). **Under Review (UR)** Stocks may be flagged as UR by the analyst, indicating that the stock's price target and/or rating are subject to possible change in the near term, usually in response to an event that may affect the investment case or valuation. **Short-Term Ratings** reflect the expected near-term (up to three months) performance of the stock and do not reflect any change in the fundamental view or investment case. **Equity Price Targets** have an investment horizon of 12 months.

EXCEPTIONS AND SPECIAL CASES: UK and European Investment Fund ratings and definitions are: **Buy:** Positive on factors such as structure, management, performance record, discount; **Neutral:** Neutral on factors such as structure, management, performance record, discount; **Sell:** Negative on factors such as structure, management, performance record, discount. **Core Banding Exceptions (CBE):** Exceptions to the standard +/-6% bands may be granted by the Investment Review Committee (IRC). Factors considered by the IRC include the stock's volatility and the credit spread of the respective company's debt. As a result, stocks deemed to be very high or low risk may be subject to higher or lower bands as they relate to the rating. When such exceptions apply, they will be identified in the Company Disclosures table in the relevant research piece.

Research analysts contributing to this report who are employed by any non-US affiliate of UBS Securities LLC are not registered/qualified as research analysts with FINRA. Such analysts may not be associated persons of UBS Securities LLC and therefore are not subject to the FINRA restrictions on communications with a subject company, public appearances, and trading securities held by a research analyst account. The name of each affiliate and analyst employed by that affiliate contributing to this report, if any, follows.

UBS Securities LLC: Aman Gulani, Jason Park, Mayur Ramdhani, Timothy Arcuri.

Company Disclosures

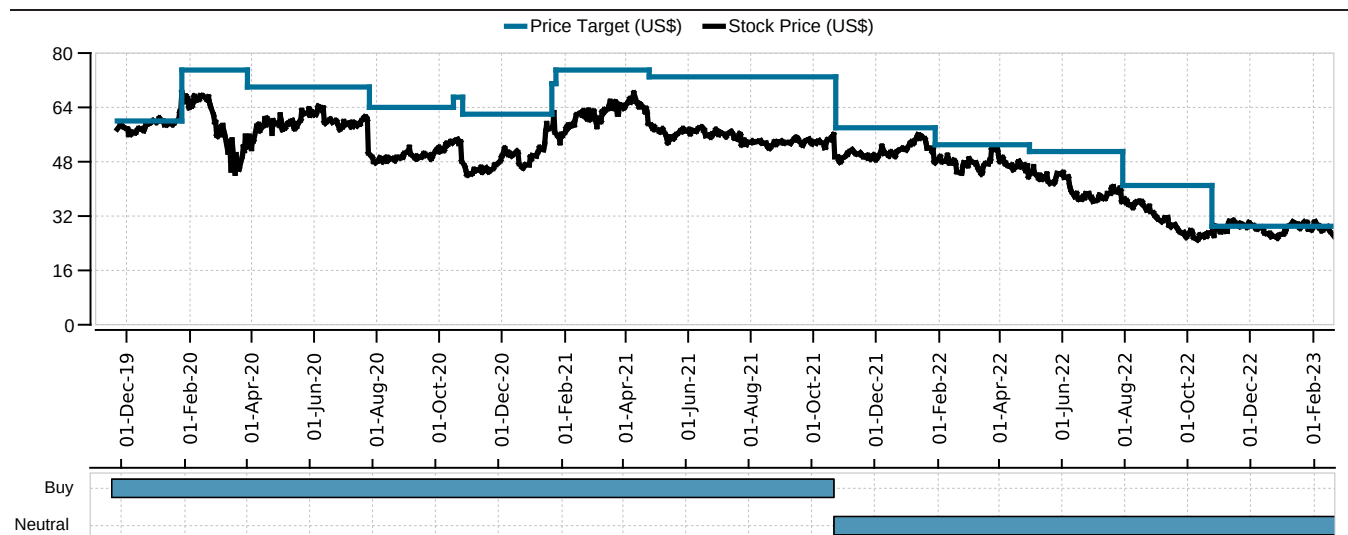
Company Name	Reuters	12-month rating	Price	Price date
Intel Corp. ^{16,6,7}	INTC.O	Neutral	US\$26.06	21 Feb 2023

Source: UBS. All prices as of local market close. Ratings in this table are the most current published ratings prior to this report. They may be more recent than the stock pricing date.

6. This company/entity is, or within the past 12 months has been, a client of UBS Securities LLC, and non-investment banking securities-related services are being, or have been, provided.
7. Within the past 12 months, UBS Securities LLC and/or its affiliates have received compensation for products and services other than investment banking services from this company/entity.
16. UBS Securities LLC makes a market in the securities and/or ADRs of this company.

Unless otherwise indicated, please refer to the Valuation and Risk sections within the body of this report. For a complete set of disclosure statements associated with the companies discussed in this report, including information on valuation and risk, please contact UBS Securities LLC, 1285 Avenue of Americas, New York, NY 10019, USA, Attention: Investment Research.

Intel Corp. (US\$)



Date	Stock Price (US\$)	Price Target (US\$)	Rating
2019-11-21	58.22	60.00	Buy
2020-01-23	63.32	75.00	Buy

Date	Stock Price (US\$)	Price Target (US\$)	Rating
2020-03-27	52.37	70.00	Buy
2020-07-24	50.59	64.00	Buy
2020-10-14	53.55	67.00	Buy
2020-10-23	48.20	62.00	Buy
2021-01-18	57.58	71.00	Buy
2021-01-22	56.66	75.00	Buy
2021-04-23	59.24	73.00	Buy
2021-10-22	49.46	58.00	Neutral
2022-01-27	48.05	53.00	Neutral
2022-04-29	43.59	51.00	Neutral
2022-07-29	36.31	41.00	Neutral
2022-10-24	27.18	29.00	Neutral

Source: UBS; as of 21-Feb-2023. All prices as of local market close. Ratings as of date shown.

The Disclaimer relevant to Global Wealth Management clients follows the Global Disclaimer.

Global Disclaimer

This document has been prepared by UBS Securities LLC, an affiliate of UBS AG. UBS AG, its subsidiaries, branches and affiliates are referred to herein as UBS.

This document is provided solely to recipients who are expressly authorized by UBS to receive it. If you are not so authorized you must immediately destroy the document.

Global Research is provided to our clients through UBS Neo, and in certain instances, UBS.com and any other system or distribution method specifically identified in one or more communications distributed through UBS Neo or UBS.com (each a system) as an approved means for distributing Global Research. It may also be made available through third party vendors and distributed by UBS and/or third parties via e-mail or alternative electronic means.

All Global Research is available on UBS Neo. Please contact your UBS sales representative if you wish to discuss your access to UBS Neo. Where Global Research refers to "UBS Evidence Lab Inside" or has made use of data provided by UBS Evidence Lab and you would like to access that data please contact your UBS sales representative. UBS Evidence Lab data is available on UBS Neo. The level and types of services provided by Global Research and UBS Evidence Lab to a client may vary depending upon various factors such as a client's individual preferences as to the frequency and manner of receiving communications, a client's risk profile and investment focus and perspective (e.g., market wide, sector specific, long-term, short-term, etc.), the size and scope of the overall client relationship with UBS and legal and regulatory constraints.

When you receive Global Research through a system, your access and/or use of such Global Research is subject to this Global Research Disclaimer and to the UBS Neo Platform Use Agreement (the "Neo Terms") together with any other relevant terms of use governing the applicable System.

When you receive Global Research via a third party vendor, e-mail or other electronic means, you agree that use shall be subject to this Global Research Disclaimer, the Neo Terms and where applicable the UBS Investment Bank terms of business (<https://www.ubs.com/global/en/investment-bank/regulatory.html>) and to UBS's Terms of Use/Disclaimer (<https://www.ubs.com/global/en/legalinfo2/disclaimer.html>). In addition, you consent to UBS processing your personal data and using cookies in accordance with our Privacy Statement (<https://www.ubs.com/global/en/legalinfo2/privacy.html>) and cookie notice (<https://www.ubs.com/global/en/legal/privacy/users.html>).

If you receive Global Research, whether through a System or by any other means, you agree that you shall not copy, revise, amend, create a derivative work, provide to any third party, or in any way commercially exploit any UBS research provided via Global Research or otherwise, and that you shall not extract data from any research or estimates provided to you via Global Research or otherwise, without the prior written consent of UBS.

In certain circumstances (including for example, if you are an academic or a member of the media) you may receive Global Research otherwise than in the capacity of a client of UBS and you understand and agree that (i) the Global Research is provided to you for information purposes only; (ii) for the purposes of receiving it you are not intended to be and will not be treated as a "client" of UBS for any legal or regulatory purpose; (iii) the Global Research must not be relied on or acted upon for any purpose; and (iv) such content is subject to the relevant disclaimers that follow.

This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject UBS to any registration or licensing requirement within such jurisdiction.

This document is a general communication and is educational in nature; it is not an advertisement nor is it a solicitation or an offer to buy or sell any financial instruments or to participate in any particular trading strategy. Nothing in this document constitutes a representation that any investment strategy or recommendation is suitable or appropriate to an investor's individual circumstances or otherwise constitutes a personal recommendation. By providing this document, none of UBS or its representatives has any responsibility or authority to provide or have provided investment advice in a fiduciary capacity or otherwise. Investments involve risks, and investors should exercise prudence and their own judgment in making their investment decisions. None of UBS or its representatives is suggesting that the recipient or any other person take a specific course of action or any action at all. The recipient should carefully read this document in its entirety and not draw inferences or conclusions from the rating alone. By receiving this document, the recipient acknowledges and agrees with the intended purpose described above and further disclaims any expectation or belief that the information constitutes investment advice to the recipient or otherwise purports to meet the investment objectives of the recipient. The financial instruments described in the document may not be eligible for sale in all jurisdictions or to certain categories of investors.

Options, structured derivative products and futures (including OTC derivatives) are not suitable for all investors. Trading in these instruments is considered risky and may be appropriate only for sophisticated investors. Prior to buying or selling an option, and for the complete risks relating to options, you must receive a copy of "The Characteristics and Risks of Standardized Options." You may read the document at <https://www.theocc.com/publications/risks/riskchap1.jsp> or ask your salesperson for a copy. Various theoretical explanations of the risks associated with these instruments have been published. Supporting documentation for any claims, comparisons, recommendations, statistics or other technical data will be supplied upon request. Past performance is not necessarily indicative of future results. Transaction costs may be significant in option strategies calling for multiple purchases and sales of options, such as spreads and straddles. Because of the importance of tax considerations to many options transactions, the investor considering options should consult with his/her tax advisor as to how taxes affect the outcome of contemplated options transactions.

Mortgage and asset-backed securities may involve a high degree of risk and may be highly volatile in response to fluctuations in interest rates or other market conditions. Foreign currency rates of exchange may adversely affect the value, price or income of any security or related instrument referred to in the document. For investment advice, trade execution or other enquiries, clients should contact their local sales representative.

The value of any investment or income may go down as well as up, and investors may not get back the full (or any) amount invested. Past performance is not necessarily a guide to future performance. Neither UBS nor any of its directors, employees or agents accepts any liability for any loss (including investment loss) or damage arising out of the use of all or any of the Information.

Prior to making any investment or financial decisions, any recipient of this document or the information should take steps to understand the risk and return of the investment and seek individualized advice from his or her personal financial, legal, tax and other professional advisors that takes into account all the particular facts and circumstances of his or her investment objectives.

Any prices stated in this document are for information purposes only and do not represent valuations for individual securities or other financial instruments. There is no representation that any transaction can or could have been effected at those prices, and any prices do not necessarily reflect UBS's internal books and records or theoretical model-based valuations and may be based on certain assumptions. Different assumptions by UBS or any other source may yield substantially different results.

No representation or warranty, either expressed or implied, is provided in relation to the accuracy, completeness or reliability of the information contained in any materials to which this document relates (the "Information"), except with respect to Information concerning UBS. The Information is not intended to be a complete statement or summary of the securities, markets or developments referred to in the document. UBS does not undertake to update or keep current the Information. Any opinions expressed in this document may change without notice and may differ or be contrary to opinions expressed by other business areas or groups, personnel or other representative of UBS. Any statements contained in this report attributed to a third party represent UBS's interpretation of the data, information and/or opinions provided by that third party either publicly or through a subscription service, and such use and interpretation have not been reviewed by the third party. In no circumstances may this document or any of the Information (including any forecast, value, index or other calculated amount ("Values")) be used for any of the following purposes:

- (i) valuation or accounting purposes;
- (ii) to determine the amounts due or payable, the price or the value of any financial instrument or financial contract; or
- (iii) to measure the performance of any financial instrument including, without limitation, for the purpose of tracking the return or performance of any Value or of defining the asset allocation of portfolio or of computing performance fees.

By receiving this document and the Information you will be deemed to represent and warrant to UBS that you will not use this document or any of the Information for any of the above purposes or otherwise rely upon this document or any of the Information.

UBS has policies and procedures, which include, without limitation, independence policies and permanent information barriers, that are intended, and upon which UBS relies, to manage potential conflicts of interest and control the flow of information within divisions of UBS and among its subsidiaries, branches and affiliates. For further information on the ways in which UBS manages conflicts and maintains independence of its research products, historical performance information and certain additional disclosures concerning UBS research recommendations, please visit <https://www.ubs.com/disclosures>.

Research will initiate, update and cease coverage solely at the discretion of UBS Research Management, which will also have sole discretion on the timing and frequency of any published research product. The analysis contained in this document is based on numerous assumptions. All material information in relation to published research reports, such as valuation methodology, risk statements, underlying assumptions (including sensitivity analysis of those assumptions), ratings history etc. as required by the Market Abuse Regulation, can be found on UBS Neo. Different assumptions could result in materially different results.

The analyst(s) responsible for the preparation of this document may interact with trading desk personnel, sales personnel and other parties for the purpose of gathering, applying and interpreting market information. UBS relies on information barriers to control the flow of information contained in one or more areas

within UBS into other areas, units, groups or affiliates of UBS. The compensation of the analyst who prepared this document is determined exclusively by research management and senior management (not including investment banking). Analyst compensation is not based on investment banking revenues; however, compensation may relate to the revenues of UBS and/or its divisions as a whole, of which investment banking, sales and trading are a part, and UBS's subsidiaries, branches and affiliates as a whole.

For financial instruments admitted to trading on an EU regulated market: UBS AG, its affiliates or subsidiaries (excluding UBS Securities LLC) acts as a market maker or liquidity provider (in accordance with the interpretation of these terms under English law or, if not carried out by UBS in the UK the law of the relevant jurisdiction in which UBS determines it carries out the activity) in the financial instruments of the issuer save that where the activity of liquidity provider is carried out in accordance with the definition given to it by the laws and regulations of any other EU jurisdictions, such information is separately disclosed in this document. For financial instruments admitted to trading on a non-EU regulated market: UBS may act as a market maker save that where this activity is carried out in the US in accordance with the definition given to it by the relevant laws and regulations, such activity will be specifically disclosed in this document. UBS may have issued a warrant the value of which is based on one or more of the financial instruments referred to in the document. UBS and its affiliates and employees may have long or short positions, trade as principal and buy and sell in instruments or derivatives identified herein; such transactions or positions may be inconsistent with the opinions expressed in this document.

Within the past 12 months UBS AG, its affiliates or subsidiaries may have received or provided investment services and activities or ancillary services as per MiFID II which may have given rise to a payment or promise of a payment in relation to these services from or to this company.

United Kingdom: This material is distributed by UBS AG, London Branch to persons who are eligible counterparties or professional clients. UBS AG, London Branch is authorised by the Prudential Regulation Authority and subject to regulation by the Financial Conduct Authority and limited regulation by the Prudential Regulation Authority. **Europe:** Except as otherwise specified herein, these materials are distributed by UBS Europe SE, a subsidiary of UBS AG, to persons who are eligible counterparties or professional clients (as detailed in the Bundesanstalt für Finanzdienstleistungsaufsicht (BaFin) Rules and according to MiFID) and are only available to such persons. The information does not apply to, and should not be relied upon by, retail clients. UBS Europe SE is authorised by the European Central Bank (ECB) and regulated by the BaFin and the ECB. **Germany, Luxembourg, the Netherlands, Belgium and Ireland:** Where an analyst of UBS Europe SE has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE. In all cases it is distributed by UBS Europe SE and UBS AG, London Branch. **Turkey:** Distributed by UBS AG, London Branch. No information in this document is provided for the purpose of offering, marketing and sale by any means of any capital market instruments and services in the Republic of Turkey. Therefore, this document may not be considered as an offer made or to be made to residents of the Republic of Turkey. UBS AG, London Branch is not licensed by the Turkish Capital Market Board under the provisions of the Capital Market Law (Law No. 6362). Accordingly, neither this document nor any other offering material related to the instruments/services may be utilized in connection with providing any capital market services to persons within the Republic of Turkey without the prior approval of the Capital Market Board. However, according to article 15 (d) (ii) of the Decree No. 32, there is no restriction on the purchase or sale of the securities abroad by residents of the Republic of Turkey. **Poland:** Distributed by UBS Europe SE (spółka z ograniczoną odpowiedzialnością) Oddział w Polsce regulated by the Polish Financial Supervision Authority. Where an analyst of UBS Europe SE (spółka z ograniczoną odpowiedzialnością) Oddział w Polsce has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE (spółka z ograniczoną odpowiedzialnością) Oddział w Polsce. **Russia:** Prepared and distributed by UBS Bank (OOO). Should not be construed as an individual Investment Recommendation for the purpose of the Russian Law - Federal Law #39-FZ ON THE SECURITIES MARKET Articles 6.1-6.2. **Switzerland:** Distributed by UBS AG to persons who are institutional investors only. UBS AG is regulated by the Swiss Financial Market Supervisory Authority (FINMA). **Italy:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, Italy Branch. Where an analyst of UBS Europe SE, Italy Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, Italy Branch. **France:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, France Branch. Where an analyst of UBS Europe SE, France Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, France Branch. **Spain:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, Spain Branch. Where an analyst of UBS Europe SE, Spain Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, Spain Branch. **Sweden:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, Sweden Branch. Where an analyst of UBS Europe SE, Sweden Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, Sweden Branch. **South Africa:** Distributed by UBS South Africa (Pty) Limited (Registration No. 1995/011140/07), an authorised user of the JSE and an authorised Financial Services Provider (FSP 7328). **Saudi Arabia:** This document has been issued by UBS AG (and/or any of its subsidiaries, branches or affiliates), a public company limited by shares, incorporated in Switzerland with its registered offices at Aeschenvorstadt 1, CH-4051 Basel and Bahnhofstrasse 45, CH-8001 Zurich. This publication has been approved by UBS Saudi Arabia (a subsidiary of UBS AG), a Saudi closed joint stock company incorporated in the Kingdom of Saudi Arabia under commercial register number 1010257812 having its registered office at Tatweer Towers, P.O. Box 75724, Riyadh 11588, Kingdom of Saudi Arabia. UBS Saudi Arabia is authorized and regulated by the Capital Market Authority to conduct securities business under license number 08113-37. **UAE / Dubai:** The information distributed by UBS AG Dubai Branch is only intended for Professional Clients and/or Market Counterparties, as classified under the DFSA rulebook. No other person should act upon this material/communication. The information is not for further distribution within the United Arab Emirates. UBS AG Dubai Branch is regulated by the DFSA in the DIFC. UBS is not licensed to provide banking services in the UAE by the Central Bank of the UAE, nor is it licensed by the UAE Securities and Commodities Authority. **Israel:** This Material is distributed by UBS AG, London Branch. UBS Securities Israel Ltd is a licensed Investment Marketmaker that is supervised by the Israel Securities Authority (ISA). UBS AG, London Branch and its affiliates incorporated outside Israel are not licensed under the Israeli Advisory Law. UBS may engage among others in issuance of Financial Assets or in distribution of Financial Assets of other issuers for fees or other benefits. UBS AG, London Branch and its affiliates may prefer various Financial Assets to which they have or may have an Affiliation (as such term is defined under the Israeli Advisory Law). Nothing in this Material should be considered as investment advice under the Israeli Advisory Law. This Material is being issued only to and/or is directed only at persons who are Eligible Clients within the meaning of the Israeli Advisory Law, and this Material must not be furnished to, relied on or acted upon by any other persons. **United States:** Distributed to US persons by either UBS Securities LLC or by UBS Financial Services Inc., subsidiaries of UBS AG; or by a group, subsidiary or affiliate of UBS AG that is not registered as a US broker-dealer (a 'non-US affiliate') to major US institutional investors only. UBS Securities LLC or UBS Financial Services Inc. accepts responsibility for the content of a report prepared by another non-US affiliate when distributed to US persons by UBS Securities LLC or UBS Financial Services Inc. All transactions by a US person in the securities mentioned in this report must be effected through UBS Securities LLC or UBS Financial Services Inc., and not through a non-US affiliate. UBS Securities LLC is not acting as a municipal advisor to any municipal entity or obligated person within the meaning of Section 15B of the Securities Exchange Act (the "Municipal Advisor Rule"), and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of the Municipal Advisor Rule. **Canada:** Distributed by UBS Securities Canada Inc., a registered investment dealer in Canada and a Member-Canadian Investor Protection Fund, or by another affiliate of UBS AG that is registered to conduct business in Canada or is otherwise exempt from registration. **Brazil:** Except as otherwise specified herein, this Material is prepared by UBS Brasil Corretora de Câmbio, Títulos e Valores Mobiliários S.A. (UBS Brasil CCTVM) to persons who are eligible investors residing in Brazil, which are considered to be Professional Investors (Investidores Profissionais), as designated by the applicable regulation, mainly the CVM Resolution No. 30 from the 11th of May 2021 (determines the duty to verify the suitability of products, services and transactions with regards to the client's profile). UBS CCTVM is a subsidiary of UBS BB Serviços de Assessoria Financeira e Participações S.A. ("UBS BB"). UBS BB is an association between UBS AG and Banco do Brasil (through its subsidiary BB – Banco de Investimentos S.A.), of which UBS AG is the majority owner and which provides investment banking services and coverage in Brazil, Argentina, Paraguay, Peru and Uruguay. **Hong Kong:** Distributed by UBS Securities Asia Limited. Please contact local licensed persons of UBS Securities Asia Limited in respect of any matters arising from, or in connection with, the analysis or document. **Singapore:** Distributed by UBS Securities Pte. Ltd. [Co. Reg. No.: 198500648C] or UBS AG, Singapore Branch. Please contact UBS Securities Pte. Ltd., an exempt financial adviser under the Singapore Financial Advisers Act (Cap. 110); or UBS AG, Singapore Branch, an exempt financial adviser under the Singapore Financial Advisers Act (Cap. 110) and a wholesale bank licensed under the Singapore Banking Act (Cap. 19) regulated by the Monetary Authority of Singapore, in respect of any matters arising from, or in connection with, the analysis or document. The recipients of this document represent and warrant that they are accredited and institutional investors as defined in the Securities and Futures Act (Cap. 289). **Japan:** Distributed by UBS Securities Japan Co., Ltd. to professional investors (except as otherwise permitted). Where this report has been prepared by UBS Securities Japan Co., Ltd., UBS Securities Japan Co., Ltd. is the author, publisher and distributor of the report. Distributed by UBS AG, Tokyo Branch to Professional Investors (except as otherwise permitted) in relation to foreign exchange and other banking businesses when relevant. **Australia:** Clients of UBS AG: Distributed by UBS AG (ABN 47 088 129 613 and holder of Australian Financial Services License No. 231087). Clients of UBS Securities Australia Ltd: Distributed by UBS Securities Australia Ltd (ABN 62 008 586 481 and holder of Australian Financial Services License No. 231098). This document contains general information and/or general advice only and does not constitute personal financial product advice. As such, the information in this document has been prepared without taking into account any investor's objectives, financial situation or needs, and investors should, before acting on the information, consider the appropriateness of the information, having regard to their objectives, financial situation and needs. If the information contained in this document relates to the acquisition, or potential acquisition of a particular financial product by a 'Retail' client as defined by section 761G of the Corporations Act 2001 where a Product Disclosure Statement would be required, the retail client should obtain and consider the Product Disclosure Statement relating to the product before making any decision about whether to acquire the product. **New Zealand:** Distributed by UBS New Zealand Ltd. UBS New Zealand Ltd is not a registered bank in New Zealand. You are being provided with this UBS publication or material because you have indicated to UBS that you are a "wholesale client" within the meaning of section 5C of the Financial Advisers Act 2008 of New Zealand (Permitted Client). This publication or material is not intended for clients who are not Permitted Clients (non-permitted Clients). If you are a non-permitted Client you must not rely on this publication or material. If despite this warning you nevertheless rely on this publication or material, you hereby (i) acknowledge that you may not rely on the content of this publication or material and that any recommendations or opinions in such this publication or material are not made or provided to you, and (ii) to the maximum extent permitted by law (a) indemnify UBS and its associates or related entities (and their respective Directors, officers, agents and Advisors) (each a 'Relevant Person') for any loss, damage, liability or claim any of them may incur or suffer as a result of, or in connection with, your unauthorised reliance on this publication or material and (b) waive any rights or remedies you may have against any Relevant Person for (or in respect of) any loss, damage, liability or claim you may incur or suffer as a result of, or in connection with, your unauthorised reliance on this publication or material. **Korea:** Distributed in Korea by UBS Securities Pte. Ltd., Seoul Branch. This report may have been edited or contributed to from time to time by affiliates of UBS Securities Pte. Ltd., Seoul Branch. This material is intended for professional/institutional clients only and not for distribution to any retail clients. **Malaysia:**

This material is authorized to be distributed in Malaysia by UBS Securities Malaysia Sdn. Bhd (Capital Markets Services License No.: CMSL/A0063/2007). This material is intended for professional/institutional clients only and not for distribution to any retail clients. **India:** Distributed by UBS Securities India Private Ltd. (Corporate Identity Number U67120MH1996PTC097299) 2/F, 2 North Avenue, Maker Maxity, Bandra Kurla Complex, Bandra (East), Mumbai (India) 400051. Phone: +912261556000. It provides brokerage services bearing SEBI Registration Number: INZ000259830; and Research Analyst services bearing SEBI Registration Number: INH000001204. It holds a SEBI Registered Merchant Banking license Number : INM000010809. UBS AG, its affiliates or subsidiaries may have debt holdings or positions in the subject Indian company/companies. UBS AG, its affiliates or subsidiaries may have financial interests (e.g. loan/derivative products, rights to or interests in investments, etc.) in the subject Indian company / companies from time to time. Within the past 12 months, UBS AG, its affiliates or subsidiaries may have received compensation for non-investment banking securities-related services and/or non-securities services from the subject Indian company/companies. The subject company/companies may have been a client/clients of UBS AG, its affiliates or subsidiaries during the 12 months preceding the date of distribution of the research report with respect to investment banking and/or non-investment banking securities-related services and/or non-securities services. With regard to information on associates, please refer to the Annual Report at: https://www.ubs.com/global/en/about_ubs/investor_relations/annualreporting.html **Taiwan:** Except as otherwise specified herein, this material may not be distributed in Taiwan. Information and material on securities/instruments that are traded in a Taiwan organized exchange is deemed to be issued and distributed by UBS Securities Pte. LTD., Taipei Branch, which is licensed and regulated by Taiwan Financial Supervisory Commission. Save for securities/instruments that are traded in a Taiwan organized exchange, this material should not constitute "recommendation" to clients or recipients in Taiwan for the covered companies or any companies mentioned in this document. No portion of the document may be reproduced or quoted by the press or any other person without authorisation from UBS. **Indonesia:** This report is being distributed by PT UBS Sekuritas Indonesia and is delivered by its licensed employee(s), including marketing/sales person, to its client. PT UBS Sekuritas Indonesia, having its registered office at Sequis Tower Level 22 unit 22-1, Jl.Jend. Sudirman, kav.71, SCBD lot 11B, Jakarta 12190. Indonesia, is a subsidiary company of UBS AG and licensed under Capital Market Law no. 8 year 1995, a holder of broker-dealer and underwriter licenses issued by the Capital Market and Financial Institution Supervisory Agency (now Otoritas Jasa Keuangan/OJK). PT UBS Sekuritas Indonesia is also a member of Indonesia Stock Exchange and supervised by Otoritas Jasa Keuangan (OJK). Neither this report nor any copy hereof may be distributed in Indonesia or to any Indonesian citizens except in compliance with applicable Indonesian capital market laws and regulations. This report is not an offer of securities in Indonesia and may not be distributed within the territory of the Republic of Indonesia or to Indonesian citizens in circumstance which constitutes an offering within the meaning of Indonesian capital market laws and regulations.

The disclosures contained in research documents produced by UBS AG, London Branch or UBS Europe SE shall be governed by and construed in accordance with English law.

UBS specifically prohibits the redistribution of this document in whole or in part without the written permission of UBS and in any event UBS accepts no liability whatsoever for any redistribution of this document or its contents or the actions of third parties in this respect. Images may depict objects or elements that are protected by third party copyright, trademarks and other intellectual property rights. © UBS 2023. The key symbol and UBS are among the registered and unregistered trademarks of UBS. All rights reserved.

Global Wealth Management Disclaimer

You receive this document in your capacity as a client of UBS Global Wealth Management. This publication has been distributed to you by UBS Switzerland AG (regulated by FINMA in Switzerland) or its affiliates ("**UBS**") with whom you have a banking relationship with. The full name of the distributing affiliate and its competent authority can be found in the country-specific disclaimer at the end of this document.

The date and time of the first dissemination of this publication is the same as the date and time of its publication.

Risk information:

You agree that you shall not copy, revise, amend, create a derivative work, provide to any third party, or in any way commercially exploit any UBS research, and that you shall not extract data from any research or estimates, without the prior written consent of UBS.

This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject UBS to any registration or licensing requirement within such jurisdiction.

This document is **for your information only**; it is not an advertisement nor is it a solicitation or an offer to buy or sell any financial instruments or to participate in any particular trading strategy. Nothing in this document constitutes a representation that any investment strategy or recommendation is suitable or appropriate to an investor's individual circumstances or otherwise constitutes a personal recommendation. By providing this document, none of UBS or its representatives has any responsibility or authority to provide or have provided investment advice in a fiduciary capacity or otherwise. Investments involve risks, and investors should exercise prudence and their own judgment in making their investment decisions. None of UBS or its representatives is suggesting that the recipient or any other person take a specific course of action or any action at all. By receiving this document, the recipient acknowledges and agrees with the intended purpose described above and further disclaims any expectation or belief that the information constitutes investment advice to the recipient or otherwise purports to meet the investment objectives of the recipient. The financial instruments described in the document may not be eligible for sale in all jurisdictions or to certain categories of investors.

Options, derivative products and futures are not suitable for all investors, and trading in these instruments is considered risky. Mortgage and asset-backed securities may involve a high degree of risk and may be highly volatile in response to fluctuations in interest rates or other market conditions. Foreign currency rates of exchange may adversely affect the value, price or income of any security or related instrument referred to in the document. For investment advice, trade execution or other enquiries, clients should contact their local sales representative.

The value of any investment or income may go down as well as up, and investors may not get back the full (or any) amount invested. Past performance is not necessarily a guide to future performance. Neither UBS nor any of its directors, employees or agents accepts any liability for any loss (including investment loss) or damage arising out of the use of all or any of the information (as defined below).

Prior to making any investment or financial decisions, any recipient of this document or the information should take steps to understand the risk and return of the investment and seek individualized advice from his or her personal financial, legal, tax and other professional advisors that takes into account all the particular facts and circumstances of his or her investment objectives.

Any prices stated in this document are for information purposes only and do not represent valuations for individual securities or other financial instruments. There is no representation that any transaction can or could have been effected at those prices, and any prices do not necessarily reflect UBS's internal books and records or theoretical model-based valuations and may be based on certain assumptions. Different assumptions by UBS or any other source may yield substantially different results.

No representation or warranty, either expressed or implied, is provided in relation to the accuracy, completeness or reliability of the information contained in any materials to which this document relates (the "Information"), except with respect to Information concerning UBS. The Information is not intended to be a complete statement or summary of the securities, markets or developments referred to in the document. UBS does not undertake to update or keep current the Information. Any opinions expressed in this document may change without notice and may differ or be contrary to opinions expressed by other business areas or groups, personnel or other representative of UBS. Any statements contained in this report attributed to a third party represent UBS's interpretation of the data, information and/or opinions provided by that third party either publicly or through a subscription service, and such use and interpretation have not been reviewed by the third party. In no circumstances may this document or any of the Information (including any forecast, value, index or other calculated amount ("Values")) be used for any of the following purposes: (i) valuation or accounting purposes; (ii) to determine the amounts due or payable, the price or the value of any financial instrument or financial contract; or (iii) to measure the performance of any financial instrument including, without limitation, for the purpose of tracking the return or performance of any Value or of defining the asset allocation of portfolio or of computing performance fees.

By receiving this document and the Information you will be deemed to represent and warrant to UBS that you will not use this document or any of the Information for any of the above purposes or otherwise rely upon this document or any of the Information.

UBS has policies and procedures, which include, without limitation, independence policies and permanent information barriers, that are intended, and upon which UBS relies, to manage potential conflicts of interest and control the flow of information within divisions of UBS (including between Global Wealth Management and UBS Global Research) and among its subsidiaries, branches and affiliates. For further information on the ways in which UBS manages conflicts and maintains independence of its research products, historical performance information and certain additional disclosures concerning UBS research recommendations, please visit <https://www.ubs.com/research-methodology>.

Research will initiate, update and cease coverage solely at the discretion of research management, which will also have sole discretion on the timing and frequency of any published research product. The analysis contained in this document is based on numerous assumptions. Different assumptions could result in materially different results.

The analyst(s) responsible for the preparation of this document may interact with trading desk personnel, sales personnel and other parties for the purpose of gathering, applying and interpreting market information. UBS relies on information barriers to control the flow of information contained in one or more areas within UBS into other areas, units, groups or affiliates of UBS. The compensation of the analyst who prepared this document is determined exclusively by research management and senior management (not including investment banking). Analyst compensation is not based on investment banking revenues; however, compensation may relate to the revenues of UBS and/or its divisions as a whole, of which investment banking, sales and trading are a part, and UBS's subsidiaries, branches and affiliates as a whole.

For financial instruments admitted to trading on an EU regulated market: UBS AG, its affiliates or subsidiaries (excluding UBS Securities LLC) acts as a market maker or liquidity provider (in accordance with the interpretation of these terms in the UK) in the financial instruments of the issuer save that where the activity of liquidity provider is carried out in accordance with the definition given to it by the laws and regulations of any other EU jurisdictions, such information is separately disclosed in this document. For financial instruments admitted to trading on a non-EU regulated market: UBS may act as a market maker save that where this activity is carried out in the US in accordance with the definition given to it by the relevant laws and regulations, such activity will be specifically disclosed in this document. UBS may have issued a warrant the value of which is based on one or more of the financial instruments referred to in the document. UBS and its affiliates and employees may have long or short positions, trade as principal and buy and sell in instruments or derivatives identified herein; such transactions or positions may be inconsistent with the opinions expressed in this document.

Options and futures are not suitable for all investors, and trading in these instruments is considered risky and may be appropriate only for sophisticated investors. Prior to buying or selling an option, and for the complete risks relating to options, you must receive a copy of "Characteristics and Risks of Standardized Options". You may read the document at <https://www.theocc.com/about/publications/character-risks.jsp> or ask your financial advisor for a copy.

Investing in structured investments involves significant risks. For a detailed discussion of the risks involved in investing in any particular structured investment, you must read the relevant offering materials for that investment. Structured investments are unsecured obligations of a particular issuer with returns linked to the performance of an underlying asset. Depending on the terms of the investment, investors could lose all or a substantial portion of their investment based on the performance of the underlying asset. Investors could also lose their entire investment if the issuer becomes insolvent. UBS does not guarantee in any way the obligations or the financial condition of any issuer or the accuracy of any financial information provided by any issuer. Structured investments are not traditional investments and investing in a structured investment is not equivalent to investing directly in the underlying asset. Structured investments may have limited or no liquidity, and investors should be prepared to hold their investment to maturity. The return of structured investments may be limited by a maximum gain, participation rate or other feature. Structured investments may include call features and, if a structured investment is called early, investors would not earn any further return and may not be able to reinvest in similar investments with similar terms. Structured investments include costs and fees which are generally embedded in the price of the investment. The tax treatment of a structured investment may be complex and may differ from a direct investment in the underlying asset. UBS and its employees do not provide tax advice. Investors should consult their own tax advisor about their own tax situation before investing in any securities.

Important Information About Sustainable Investing Strategies: Sustainable investing strategies aim to consider and incorporate environmental, social and governance (ESG) factors into investment process and portfolio construction. Strategies across geographies and styles approach ESG analysis and incorporate the findings in a variety of ways. Incorporating ESG factors or Sustainable Investing considerations may inhibit the portfolio manager's ability to participate in certain investment opportunities that otherwise would be consistent with its investment objective and other principal investment strategies. The returns on a portfolio consisting primarily of sustainable investments may be lower or higher than portfolios where ESG factors, exclusions, or other sustainability issues are not considered by the portfolio manager, and the investment opportunities available to such portfolios may differ. Companies may not necessarily meet high performance standards on all aspects of ESG or sustainable investing issues; there is also no guarantee that any company will meet expectations in connection with corporate responsibility, sustainability, and/or impact performance.

Within the past 12 months UBS Switzerland AG, its affiliates or subsidiaries may have received or provided investment services and activities or ancillary services as per MiFID II which may have given rise to a payment or promise of a payment in relation to these services from or to this company.

If you require detailed information on disclosures of interest or conflict of interest as required by Market Abuse Regulation please contact the mailbox MAR_disclosures_twopager@ubs.com. Please note that e-mail communication is unsecured.

External Asset Managers / External Financial Consultants: In case this research or publication is provided to an External Asset Manager or an External Financial Consultant, UBS expressly prohibits that it is redistributed by the External Asset Manager or the External Financial Consultant and is made available to their clients and/or third parties.

Bahrain: UBS is a Swiss bank not licensed, supervised or regulated in Bahrain by the Central Bank of Bahrain to undertake banking or investment business activities in Bahrain. Therefore, prospects/clients do not have any protection under local banking and investment services laws and regulations. **Brazil:** This report is only intended for Brazilian residents who are directly purchasing or selling securities in the Brazil capital market through a local authorized institution. **Canada:** The information contained herein is not, and under no circumstances is to be construed as, a prospectus, an advertisement, a public offering, an offer to sell securities described herein, solicitation of an offer to buy securities described herein, in Canada or any province or territory thereof. Any offer or sale of the securities described herein in Canada will be made only under an exemption from the requirements to file a prospectus with the relevant Canadian securities regulators and only by a dealer properly registered under applicable securities laws or, alternatively, pursuant to an exemption from the dealer registration requirement in the relevant province or territory of Canada in which such offer or sale is made. Under no circumstances is the information contained herein to be construed as investment advice in any province or territory of Canada and is not tailored to the needs of the recipient. To the extent that the information contained herein references securities of an issuer incorporated, formed or created under the laws of Canada or a province or territory of Canada, any trades in such securities must be conducted through a dealer registered in Canada or, alternatively, pursuant to a dealer registration exemption. No securities commission or similar regulatory authority in Canada has reviewed or in any way passed upon these materials, the information contained herein or the merits of the securities described herein and any representation to the contrary is an offence. In Canada, this publication is distributed by UBS Investment Management Canada Inc. **China:** This report and any offering material such as term sheet, research report, other product or service documentation or any other information (the "Material") sent with this report was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the material erroneously, UBS asks that you kindly delete it and inform UBS immediately. This report is prepared by UBS Switzerland AG or its offshore subsidiary or affiliate (collectively as "UBS Offshore"). UBS Offshore is an entity incorporated out of China and is not licensed, supervised or regulated in China to carry out banking or securities business. The recipient should not contact the analysts or UBS Offshore which produced this report for advice as they are not licensed to provide securities investment advice in China. UBS Investment Bank (including Research) has its own wholly independent research and views which at times may vary from the views of UBS Global Wealth Management. The recipient should not use this document or otherwise rely on any of the information contained in this report in making investment decisions and UBS takes no responsibility in this regard. **Czech Republic:** UBS is not a licensed bank in the Czech Republic and thus is not allowed to provide regulated banking or investment services in the Czech Republic. This communication and/or material is distributed for marketing purposes and constitutes a "Commercial Message" under the laws of Czech Republic in relation to banking and/or investment services. Please notify UBS if you do not wish to receive any further correspondence. **Denmark:** This publication is not intended to constitute a public offer under Danish law. It is distributed only for information purposes to clients of UBS Europe SE, Denmark Branch, filial af UBS Europe SE, with place of business at Sankt Annae Plads 13, 1250 Copenhagen, Denmark, registered with the Danish Commerce and Companies Agency, under No. 38 17 24 33. UBS Europe SE, Denmark Branch, filial af UBS Europe SE is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the Danish Financial Supervisory Authority (Finanstilsynet), to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Egypt:** Securities or other investment products are not being offered or sold by UBS to the public in Egypt and they have not been and will not be registered with the Egyptian Financial Regulatory Authority (FRA). **France:** This publication is not intended to constitute a public offer under French law, it does not constitute a personal recommendation as it is distributed only for information purposes to clients of UBS (France) S.A., French "société anonyme" with share capital of € 132.975.556, at 69 boulevard Haussmann F-75008 Paris, registered with the "Registre du Commerce et des Sociétés" of Paris under N° B 421 255 670. UBS (France) S.A. is a provider of investment services duly authorized according to the terms of the "Code monétaire et financier", regulated by French banking and financial authorities as the "Autorité de contrôle prudentiel et de résolution" and "Autorité des marchés financiers", to which this publication has not been submitted for approval. **Germany:** This publication is not intended to constitute a public offer under German law. It is distributed only for information purposes to clients of UBS Europe SE, Germany, with place of business at Bockenheimer Landstrasse 2-4, 60306 Frankfurt am Main. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin") and supervised jointly by the European Central Bank, the German Central Bank (Deutsche Bundesbank) and BaFin, to which this publication has not been submitted for approval. **Hong Kong SAR:** This publication is distributed to clients of UBS AG Hong Kong Branch by UBS AG Hong Kong Branch, a licensed bank under the Hong Kong Banking Ordinance and a registered institution under the Securities and Futures Ordinance. UBS AG Hong Kong Branch is incorporated in Switzerland with limited liability. **India:** Distributed by UBS Securities India Private Ltd. (Corporate Identity Number U67120MH1996PTC097299) 2/F, 2 North Avenue, Maker Maxity, Bandra Kurla Complex, Bandra (East), Mumbai (India) 400051. Phone: +912261556000. It provides brokerage services bearing SEBI Registration Number INZ000259830; merchant banking services bearing SEBI Registration Number: INM000010809 and Research Analyst services bearing SEBI Registration Number: INH000001204. UBS AG, its affiliates or subsidiaries may have debt holdings or positions in the subject Indian company/companies. UBS AG, its affiliates or subsidiaries may have financial interests (e.g. like loan/derivative products, rights to or interests in investments, etc.) in the subject Indian company/companies from time to time. Within the past 12 months, UBS AG, its affiliates or subsidiaries may have received compensation for non-investment banking securities-related services and/or non-securities services from the subject Indian company/companies. The subject company/companies may have been a client/clients of UBS AG, its affiliates or subsidiaries during the 12 months preceding the date of distribution of the research report with respect to investment banking and/or non-investment banking securities-related services and/or non-securities services. With regard to information on associates, please refer to the Annual Report at: https://www.ubs.com/global/en/about_ubs/investor_relations/annualreporting.html. **Indonesia:** This communication and any offering material term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete the e-mail and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed without the consent of UBS. None of the Material has been registered or filed under the prevailing laws and with any financial or regulatory authority in your jurisdiction. The Material may not have been approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or products or futures contracts). The Material is neither an offer nor a solicitation to

enter into any transaction or contract (including futures contracts) nor is it an offer to buy or to sell any securities or products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you.

Israel: UBS is a premier global financial firm offering wealth management, asset management and investment banking services from its headquarters in Switzerland and its operations in over 50 countries worldwide to individual, corporate and institutional investors. In Israel, UBS Switzerland AG is registered as Foreign Dealer in cooperation with UBS Wealth Management Israel Ltd., a wholly owned UBS subsidiary. UBS Wealth Management Israel Ltd. is an Investment Marketing licensee which engages in Investment Marketing and is regulated by the Israeli Securities Authority. This publication is intended for information only and is not intended as an offer to buy or solicitation of an offer. Furthermore, this publication is not intended as an investment advice. No action has been, or will be, taken in Israel that would permit an offering of the product(s) mentioned in this document or a distribution of this document to the public in Israel. In particular, this document has not been reviewed or approved by the Israeli Securities Authority. The product(s) mentioned in this document is/are being offered to a limited number of sophisticated investors who qualify as one of the investors listed in the first supplement to the Israeli Securities Law, 5728-1968. This document may not be reproduced or used for any other purpose, nor be furnished to any other person other than those to whom copies have been sent. Anyone who purchases the product(s) mentioned herein shall do so for its own benefit and for its own account and not with the aim or intention of distributing or offering the product(s) to other parties. Anyone who purchases the product(s) shall do so in accordance with its own understanding and discretion and after it has received any relevant financial, legal, business, tax or other advice or opinion required by it in connection with such purchase(s). The word "advice" and/or any of its equivalent terms shall be read and construed in conjunction with the definition of the term "investment marketing" as defined under the Israeli Regulation of Investment Advice, Investment Marketing and Portfolio Management Law. The Swiss laws and regulations require a number of mandatory disclosures to be made in independent financial research reports or recommendations. Pursuant to the Swiss Financial Market Infrastructure Act and the Financial Market Infrastructure Ordinance-FINMA, banks must disclose the percentage of voting rights they hold in companies being researched, if these holdings are equal to or exceed the statutory thresholds. In addition, the Directives on the Independence of Financial Research, issued by the Swiss Bankers Association, mandate a number of disclosures, including the disclosure of potential conflicts of interest, the participation within previous 12 months in any securities issues on behalf of the company being researched, as well as the fact that remuneration paid to the financial analysts is based generally upon the performance of (i) the new issues department or investment banking; or (ii) securities trading performance (including proprietary trading) or sales.

Italy: This publication is not intended to constitute a public offer under Italian law. It is distributed only for information purposes to clients of UBS Europe SE, Succursale Italia, with place of business at Via del Vecchio Politecnico, 3-20121 Milano. UBS Europe SE, Succursale Italia is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the Bank of Italy (Banca d'Italia) and the Italian Financial Markets Supervisory Authority (CONSOB - Commissione Nazionale per le Società e la Borsa), to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin.

Jersey: UBS AG, Jersey Branch, is regulated and authorized by the Jersey Financial Services Commission for the conduct of banking, funds and investment business. Where services are provided from outside Jersey, they will not be covered by the Jersey regulatory regime. UBS AG, Jersey Branch is a branch of UBS AG a public company limited by shares, incorporated in Switzerland whose registered offices are at Aeschenvorstadt 1, CH-4051 Basel and Bahnhofstrasse 45, CH 8001 Zurich. UBS AG, Jersey Branch's principal place of business is 1, IFC Jersey, Jersey, JE2 3BX.

Luxembourg: This publication is not intended to constitute a public offer under Luxembourg law. It is distributed only for information purposes to clients of UBS Europe SE, Luxembourg Branch ("UBS Luxembourg"), R.C.S. Luxembourg n° B209123, with registered office at 33A, Avenue J. F. Kennedy, L-1855 Luxembourg. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea (HRB n° 107046), with registered office at Bockenheimer Landstrasse 2-4, D-60306 Frankfurt am Main, Germany, duly authorized by the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin") and subject to the joint prudential supervision of BaFin, the European Central Bank and the central bank of Germany (Deutsche Bundesbank). UBS Luxembourg is furthermore supervised by the Luxembourg prudential supervisory authority (Commission de Surveillance du Secteur Financier), in its role as host member state authority. This publication has not been submitted for approval to any public supervisory authority.

Malaysia: This communication and any offering material term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete the e-mail and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or derivatives products). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or derivatives products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you.

Mexico: This information is distributed by UBS Asesores México, S.A. de C.V. ("UBS Asesores"), an affiliate of UBS Switzerland AG, incorporated as a non-independent investment advisor under the Mexican regulation due to the relation with a Foreign Bank. UBS Asesores is registered under number 30060-001-(14115)-21/06/2016 and subject to the supervision of the Mexican Banking and Securities Commission ("CNBV") exclusively regarding the rendering of (i) portfolio management services, (ii) securities investment advisory services, analysis and issuance of individual investment recommendations, and (iii) anti-money laundering and terrorism financing matters. This UBS publication or any material related thereto is addressed only to Sophisticated or Institutional Investors located in Mexico. Research reports only reflect the views of the analysts responsible for the report. The compensation of the analyst(s) who prepared this report is determined exclusively by research management and senior management of any entity of UBS Group to which such analyst(s) render(s) services.

Nigeria: The investment products mentioned in this material are not being offered or sold by UBS to the public in Nigeria and they have not been submitted for approval nor registered with the Securities and Exchange Commission of Nigeria. If you are interested in products of this nature, please let us know. The investment products mentioned in this material are not being directed to, and are not being made available for subscription by any persons within Nigeria other than the selected investors to whom the offer materials have been addressed as a private sale or domestic concern within the exemption and meaning of Section 69(2) of the Investments and Securities Act, 2007 (ISA). This material has been provided to you at your specific unsolicited request and for your information only.

Philippines: This communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete the e-mail and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or derivatives products). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or derivatives products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you.

Portugal: UBS Switzerland AG is not licensed to conduct banking and financial activities in Portugal nor is UBS Switzerland AG supervised by the Portuguese regulators (Bank of Portugal "Banco de Portugal" and Portuguese Securities Exchange Commission "Comissão do Mercado de Valores Mobiliários").

Qatar: UBS Qatar LLC is authorized by the Qatar Financial Centre Regulatory Authority, with QFC no. 01169, and has its registered office at 14th Floor, Burj Alfardan Tower, Building 157, Street No. 301, Area No. 69, Al Majdumi, Lusail, Qatar. UBS Qatar LLC neither offers any brokerage services nor executes any order with, for or on behalf of its clients. A client order will have to be placed with, and executed by, UBS Switzerland AG in Switzerland or an affiliate of UBS Switzerland AG, that is domiciled outside Qatar. It is in the sole discretion of UBS Switzerland AG in Switzerland or its affiliate to accept or reject an order and UBS Qatar LLC does not have authority to provide a confirmation in this respect. UBS Qatar LLC may however communicate payment orders and investment instructions to UBS Switzerland AG in Switzerland for receipt, acceptance and execution. UBS Qatar LLC is not authorized to act for and on behalf of UBS Switzerland AG or an affiliate of UBS Switzerland AG. This document and any attachments hereto are intended for eligible counterparties and business customers only.

Russia: This document or information contained therein is for information purposes only and constitutes neither a public nor a private offering, is not an invitation to make offers, to sell, exchange or otherwise transfer any financial instruments in the Russian Federation to or for the benefit of any Russian person or entity and does not constitute an advertisement or offering of securities in the Russian Federation within the meaning of Russian securities laws. The information contained herein is not an "individual investment recommendation" as defined in Federal Law of 22 April 1996 No 39-FZ "On Securities Market" (as amended) and the financial instruments and operations specified herein may not be suitable for your investment profile or your investment goals or expectations. The determination of whether or not such financial instruments and operations are in your interests or are suitable for your investment goals, investment horizon or the acceptable risk level is your responsibility. We assume no liability for any losses connected with making any such operations or investing into any such financial instruments and we do not recommend to use such information as the only source of information for making an investment decision.

Saudi Arabia: UBS Saudi Arabia is authorised and regulated by the Capital Market Authority to conduct securities business under licence number 08113-37.

Singapore: Clients of UBS AG Singapore branch are asked to please contact UBS AG Singapore branch, an exempt financial adviser under the Singapore Financial Advisers Act (Cap. 110) and a wholesale bank licensed under the Singapore Banking Act (Cap. 19) regulated by the Monetary Authority of Singapore, in respect of any matters arising from, or in connection with, the analysis or report. UBS AG is incorporated in Switzerland with limited liability. UBS AG has a branch registered in Singapore (UEN S98FC5560C). This communication and any offering material term sheet, research report, other product or service documentation or any other

information (the "Material") sent with this communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete the e-mail and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved or endorsed by any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or products or futures contracts). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you.

Sweden: This publication is not intended to constitute a public offer under Swedish law. It is distributed only for information purposes to clients of UBS Europe SE, Sweden Bankfilial, with place of business at Regeringsgatan 38, 11153 Stockholm, Sweden, registered with the Swedish Companies Registration Office under Reg. No 516406-1011. UBS Europe SE, Sweden Bankfilial is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the Swedish supervisory authority (Finansinspektionen), to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Taiwan:** This material is provided by UBS AG, Taipei Branch in accordance with laws of Taiwan, in agreement with or at the request of clients/prospects. **Thailand:** This communication and any offering material, term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication were done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete the e-mail and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or derivatives products). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or derivatives products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Turkey:** The information in this document is not provided for the purpose of offering, marketing or sale of any capital market instrument or service in the Republic of Turkey. Therefore, this document may not be considered as an offer made, or to be made, to residents of the Republic of Turkey in the Republic of Turkey. UBS Switzerland AG is not licensed by the Turkish Capital Market Board (the CMB) under the provisions of the Capital Market Law (Law No. 6362). Accordingly, neither this document nor any other offering material related to the instrument/service may be utilized in connection with providing any capital market services to persons within the Republic of Turkey without the prior approval of the CMB. However, according to article 15 (d) (ii) of the Decree No. 32 residents of the Republic of Turkey are allowed to purchase or sell the financial instruments traded in financial markets outside of the Republic of Turkey. Further to this, pursuant to article 9 of the Communiqué on Principles Regarding Investment Services, Activities and Ancillary Services No. III-37.1, investment services provided abroad to residents of the Republic of Turkey based on their own initiative are not restricted. **United Arab Emirates (UAE):** UBS is not a financial institution licensed in the UAE by the Central Bank of the UAE nor by the Emirates' Securities and Commodities Authority and does not undertake banking activities in the UAE. UBS AG Dubai Branch is licensed by the DFSA in the DIFC. This document is provided for your information only and does not constitute financial advice. **United Kingdom:** This document is issued by UBS Wealth Management, a division of UBS AG which is authorised and regulated by the Financial Market Supervisory Authority in Switzerland. In the United Kingdom, UBS AG is authorised by the Prudential Regulation Authority and is subject to regulation by the Financial Conduct Authority and limited regulation by the Prudential Regulation Authority. Details about the extent of regulation by the Prudential Regulation Authority are available from us on request. A member of the London Stock Exchange. This publication is distributed to retail clients of UBS Wealth Management. **Ukraine:** UBS is not registered and licensed as a bank/financial institution under Ukrainian legislation and does not provide banking and other financial services in Ukraine. UBS has not made, and will not make, any offer of the mentioned products to the public in Ukraine. No action has been taken to authorize an offer of the mentioned products to the public in Ukraine and the distribution of this document shall not constitute financial services for the purposes of the Law of Ukraine "On Financial Services and State Regulation of Financial Services Markets" dated 12 July 2001. Any offer of the mentioned products shall not constitute an investment advice, public offer, circulation, transfer, safekeeping, holding or custody of securities in the territory of Ukraine. Accordingly, nothing in this document or any other document, information or communication related to the mentioned products shall be interpreted as containing an offer, a public offer or invitation to offer or to a public offer, or solicitation of securities in the territory of Ukraine or investment advice under Ukrainian law. Electronic communication must not be considered as an offer to enter into an electronic agreement or other electronic instrument within the meaning of the Law of Ukraine "On Electronic Commerce" dated 3 September 2015. This document is strictly for private use by its holder and may not be passed on to third parties or otherwise publicly distributed. **USA:** Distributed to US persons by UBS Financial Services Inc. or UBS Securities LLC, subsidiaries of UBS AG. UBS Switzerland AG, UBS Europe SE, UBS Bank, S.A., UBS Brasil Administradora de Valores Mobiliários Ltda., UBS Asesores México, S.A. de C.V., UBS SuMi TRUST Wealth Management Co., Ltd., UBS Wealth Management Israel Ltd. and UBS Menkul Degerler AS are affiliates of UBS AG. **UBS Financial Services Inc. accepts responsibility for the content of a report prepared by a non-US affiliate when it distributes reports to US persons. All transactions by a US person in the securities mentioned in this report should be effected through a US-registered broker dealer affiliated with UBS, and not through a non-US affiliate. The contents of this report have not been and will not be approved by any securities or investment authority in the United States or elsewhere. UBS Financial Services Inc. is not acting as a municipal advisor to any municipal entity or obligated person within the meaning of Section 15B of the Securities Exchange Act (the "Municipal Advisor Rule") and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of the Municipal Advisor Rule.**

© UBS 2023. The key symbol and UBS are among the registered and unregistered trademarks of UBS. All rights reserved.

